

CANDIDATE'S NAME:

SIGNATURE

<i>Personal number</i>							

P530/1
Biology
Paper 1
2 $\frac{1}{2}$ hours



Apex Examination Council

PRE-ENTRY EXAMINATIONS SET TWO 2026

Uganda Advanced certificate of education

Biology

Paper 1 (Theory)

2 hours 30 minutes

INSTRUCTIONS TO CANDIDATES

- This paper consists of two sections, A and B.
- Section A contains two compulsory items.
- Section B contains three items.
- Answer ONE item from Section B.
- Answer THREE items in all.
- All responses should be based on biological principles and evidence provided.

SECTION A

Answer all items

ITEM 1

A commercial rabbit farm in Kayunga District experienced sudden deaths among rabbits shortly after they were fed wilted cassava leaves. Within minutes, some rabbits developed rapid breathing, muscle weakness and loss of coordination. Several animals collapsed and died despite showing no signs of airway obstruction.

Veterinary investigations revealed the following results.

Table 1

Observation	Control rabbit	Poisoned rabbit
Breathing rate (breaths/min)	45	110
Venous blood colour	Dark red	Bright red
Mitochondrial ATP (umol/mg protein)	5.5	0.2
NADH concentration (umol/g tissue)	0.3	2.1
Tissue damage severity	None	High brain and heart
Survival rate (%)	100	40

Further laboratory analysis confirmed cyanide poisoning.

Tasks

- Explain how cyanide interferes with aerobic respiration in cells.
- Account for the high NADH concentration and low ATP production in affected rabbits.
- Explain why brain and heart tissues suffered the greatest damage.
- Suggest measures that farmers can use to prevent similar incidents in livestock production.

ITEM 2

Napak District experiences high temperatures, low rainfall, shallow soils and limited soil organic matter. Despite these harsh conditions, some indigenous crop species continue to produce reasonable yields while introduced species often perform poorly.

Agricultural extension workers conducted a comparative study and obtained the following data.

Table 2

Condition	Native plants	Non-native plants
The type of leaf anatomy	Kranz anatomy Present	Kranz anatomy Absent
Distribution of stomata	Many stomata on lower leaf epidermis	Many stomata on upper leaf epidermis
Size of leaf cuticle	Thick thin	Thin
Number of hairs on the leaf	Numerous	Very few

Tasks

- Explain how each adaptation shown by the native plants increases survival under the prevailing environmental conditions.
- Distinguish between C₃ and C₄ photosynthetic pathways.
- Account for the differences in yield between native and non-native crops.
- Recommend strategies that can improve productivity and resilience of crops in the district.

SECTION B**Answer only one item****ITEM 3**

During a geography field study in Karamoja, students camped in a semi-desert environment where daytime temperatures reached 39°C and night temperatures fell to 15°C.

A group of students forgot to carry sufficient drinking water and protective shade. By the second day, some developed dry lips, dizziness, reduced urine output and elevated body temperature. The school nurse collected the following data.

Table 3

Students	Body temp (°C)	Urine volume ml/hr	Pulse Beats/mins	Plasma ADH level (pg/ml)
A(no shade no water)	39.8	8	120	20
B(same as A)	39.2	10	118	22
C(with caffeine)	37.5	40	85	6
D(in shade hydrated)	36.8	60	75	5

Tasks

- (a) Analyse how the physiological responses observed demonstrate homeostatic regulation of body temperature and water balance.
- (b) Explain the role of ADH in maintaining fluid balance.
- (c) Account for the differences observed between Student C and Students A and B.
- (d) Suggest scientifically sound measures for preventing dehydration and overheating during future field activities.

OR

ITEM 4

A district health team conducted a genetic screening programme to determine the inheritance pattern of sickle-cell trait in a population around Lake Kyoga. The results indicated increasing awareness of genetic disorders but also revealed limited understanding of inheritance patterns among community members.

A family study showed the following parental genotypes:

Father: HbA HbS

Mother: HbA HbS

Parameter	Alex	Chris
Pulse rate after race (beat/min)	115	155
Blood oxygen saturation (%)	93	85
Breathing rate (breaths/min)	18	28
Stroke volume (ml/beat)	120	100

Tasks

- (a) Using a genetic diagram, determine the expected genotypes and phenotypes of the offspring.
- (b) Calculate the probability of producing:
 - (i) a normal child,
 - (ii) a carrier child,
 - (iii) a child with sickle-cell disease.
- (c) Explain the importance of genetic counselling in communities affected by inherited disorders.
- (d) Assess the factors that influence changes in allele frequencies within populations over time.

OR

ITEM 5

A conservation study carried out in Queen Elizabeth National Park indicated a decline in several animal populations over the last decade. Researchers identified habitat destruction, climate change, poaching and invasive species as major threats.

The study also revealed that some species were becoming increasingly vulnerable to extinction due to reduced genetic diversity and environmental changes.

Tasks

- (a) Explain the ecological factors responsible for the decline in biodiversity.
- (b) Assess the impact of climate change on ecosystem stability.
- (c) Explain how natural selection contributes to adaptation and survival of species.
- (d) Suggest strategies that can be used to conserve biodiversity and maintain ecological balance.

END

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